

# T3 final-EFT one-loop RGE term comparison: EFT\_2\_after\_S1\_S2

All T3 particles have been integrated out at this stage. The active theory is the SM plus the dimension-five Weinberg operator. Rows are the UV model configurations that produced the matched coefficient, while columns align the additive structures in its one-loop beta function. Each populated cell contains the complete term, including its coefficient and the factor  $C_5$ . We use  $16\pi^2 \mu dC_5/d\mu = \beta_{C_5}^{(1)}$ .

## Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1$                     | $D_\mu \supset ig_1 Y B_\mu$                                       |
| $g_2$                     | $D_\mu \supset ig_2 T^A W_\mu^A$                                   |
| $g_3$                     | $D_\mu \supset ig_3 t^A G_\mu^A$                                   |
| $Y_u$                     | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$                           |
| $Y_d$                     | $-\bar{Q} Y_d H d + \text{h.c.}$                                   |
| $Y_e$                     | $-\bar{L} Y_e H e + \text{h.c.}$                                   |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F} M_F F$                                                   |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2$                   | $m_1^2 S_1^\dagger S_1$                                            |
| $m_2^2$                   | $m_2^2 S_2^\dagger S_2$                                            |
| $\lambda_1$               | $\frac{1}{2} \lambda_1 (H^\dagger H)^2$                            |
| $\lambda_{S_1}^{(1)}$     | $\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$              |
| $\lambda_{S_2}^{(1)}$     | $\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$              |
| $\lambda_{HS_1}^{(1)}$    | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$              |
| $\lambda_{HS_2}^{(1)}$    | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$              |
| $\lambda_{HS_1}^{(A)}$    | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$      |
| $\lambda_{HS_2}^{(A)}$    | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$      |
| $\lambda_5$               | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$                      |
| $\lambda_{12}^{(1)}$      | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$            |
| $\lambda_{12}^{(A)}$      | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$    |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$       |
| $\lambda_{S_1}^{(A)}$     | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$   |
| $\lambda_{S_2}^{(A)}$     | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$   |
| $\alpha$                  | $Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$   |
| $C_2(d)$                  | $(d^2 - 1)/4$                                                      |

|                        |                                                      |
|------------------------|------------------------------------------------------|
| $T(d)$                 | $d(d^2-1)/12$                                        |
| $C_{LLS_1S_2}^{ij;AB}$ | $L_iL_jS_1^AS_2^B$                                   |
| $C_{ijab}$             | $L_iL_j\phi_a\phi_b$                                 |
| $C_5^{ij}$             | $(L_i^TCL_j)HH$                                      |
| $R_W$                  | $16\pi^2\beta_{C_5}\supset R_W\lambda_5C_{LLS_1S_2}$ |

$$\beta_{C_5}^{(1)}$$

**Quantity:**  $C_5$

$$16\pi^2\,\mu\frac{dC_5}{d\mu}=\beta_{C_5}^{(1)}$$

**Terms 1 of 2**

| Column Representative term structure |                     |
|--------------------------------------|---------------------|
| T1                                   | $C_5g_2^2$          |
| T2                                   | $C_5\lambda_1$      |
| T3                                   | $C_5Y_eY_e^\dagger$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1           | T2              | T3                   |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------|-----------------|----------------------|
| T3-C  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 3     | 1     | $-3C_5g_2^2$ | $2C_5\lambda_1$ | $-C_5Y_eY_e^\dagger$ |

Quantity:  $C_5$

Terms 2 of 2

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $C_5 Y_d Y_d^\dagger$         |
| T2     | $C_5 Y_u Y_u^\dagger$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                     | T2                     |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|------------------------|------------------------|
| T3-C  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 3     | 1     | $6C_5 Y_d Y_d^\dagger$ | $6C_5 Y_u Y_u^\dagger$ |